

Film reference: *Mission to Mars* (Brian De Palma, Touchstone Pictures 2000)

Concept: The time required for air to escape from a spaceship depends on many factors, including the size of the hole through which the air will escape and the volume of the spaceship.

Web references for the International Space Station:

- [Article from The Gaurdian about the discovery and repair of the air leak](#)
- [NASA.gov news about the ISS for August 2018](#)
- [Facts and Figures for the ISS](#)

In Section 3.4.1 of the text (*Exploring Science Through Science Fiction, 2nd Ed.*), the reader is guided through the steps of a calculation of the time required for air to leak out of a large, interplanetary spaceship through a hole roughly the diameter of a finger. The scenario is portrayed in the 2000 movie *Mission to Mars*. The result of the calculation turns out to be many hours, not just a few minutes, as the movie suggests.

On August 30, 2018, an air leak was detected aboard the International Space Station (ISS). The leak was traced to a small drill hole in a Soyuz space capsule, docked with the ISS.

Problem:

Use the example calculation worked out in the text to come up with an estimate of the time required for half the air to leak out of the International Space Station through the hole shown in the photograph in the Guardian article (reference above).

- a. Go to the official NASA website for Facts and Figures for the International Space Station and find the “total pressurized volume”, V , of the ISS. Express the value in cubic meters.
- b. Consult the recommended web articles (the Guardian and the official NASA news) to find an estimate of the diameter of the hole that caused the air leak. (Note: The diameter will probably be given in units of millimeters. Be sure to convert this to meters. Calculate the area of the hole, A , in units of square meters.)
- c. Using the example calculation in the text (and shown in class) estimate the time required for half of the air to leak out of the ISS.